

4.2. Neka $u = \begin{pmatrix} u_1 \\ u_2 \end{pmatrix}$, $v = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix}$, $w = \begin{pmatrix} w_1 \\ w_2 \end{pmatrix}$

Osnovni:

$$\langle u, v \rangle = \begin{pmatrix} u_1 & u_2 \end{pmatrix} \begin{pmatrix} a & b \\ b & c \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix}$$

Pretpostavimo implicitno $a_{11} > 0$ i $\det(A) > 0 \Rightarrow \langle, \rangle$ je skalarni proizvod na $\mathbb{R}^2$

Proizvodnja:

$$\begin{pmatrix} u_1 & u_2 \end{pmatrix} \begin{pmatrix} a & b \\ b & c \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = a u_1 v_1 + b u_1 v_2 + b u_2 v_1 + c u_2 v_2$$

$$\begin{pmatrix} v_1 & v_2 \end{pmatrix} \begin{pmatrix} a & b \\ b & c \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = a v_1 u_1 + b v_1 u_2 + b v_2 u_1 + c v_2 u_2 = a u_1 v_1 + b u_1 v_2 + b u_2 v_1 + c u_2 v_2$$

Dakle, ako, $\langle u, v \rangle = \langle v, u \rangle$

Proizvodnja:

$$\langle u, v \rangle = \begin{pmatrix} u_1 & u_2 \end{pmatrix} \begin{pmatrix} a & b \\ b & c \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = a u_1 v_1 + b u_1 v_2 + b u_2 v_1 + c u_2 v_2 = \langle a u_1 v_1 + b u_1 v_2 + b u_2 v_1 + c u_2 v_2 \rangle = \langle a u, v \rangle$$

$$\langle u, v, w \rangle = \begin{pmatrix} u_1 & u_2 \end{pmatrix} \begin{pmatrix} a & b \\ b & c \end{pmatrix} \begin{pmatrix} v_1 + w_1 \\ v_2 + w_2 \end{pmatrix} = a u_1 (v_1 + w_1) + b u_1 (v_2 + w_2) + b u_2 (v_1 + w_1) + c u_2 (v_2 + w_2) = a u_1 v_1 + b u_1 v_2 + b u_2 v_1 + c u_2 v_2 + a u_1 w_1 + b u_1 w_2 + b u_2 w_1 + c u_2 w_2 = \langle u, v \rangle + \langle u, w \rangle$$

Proizvodnja definicija:

$$\langle u, u \rangle = \begin{pmatrix} u_1 & u_2 \end{pmatrix} \begin{pmatrix} a & b \\ b & c \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = a u_1^2 + b u_1 u_2 + b u_2 u_1 + c u_2^2 = a u_1^2 + 2 b u_1 u_2 + c u_2^2$$

Osnovni na osnovu:

$$a u_1^2 + 2 b u_1 u_2 + c u_2^2 = a \left(u_1 + \frac{b}{a} u_2 \right)^2 + \left(c - \frac{b^2}{a} \right) u_2^2 = a \left(u_1 + \frac{b}{a} u_2 \right)^2 + \left(\frac{c a - b^2}{a} \right) u_2^2$$

Neka $\det(A) = ac - b^2$:

$$a \left(u_1 + \frac{b}{a} u_2 \right)^2 + \left(\frac{\det(A)}{a} \right) u_2^2$$

Na osnovu ovoga, ako $a > 0$ i $\det(A) > 0$, tada je $\frac{\det(A)}{a} > 0$.

Tada:

$$a \left(u_1 + \frac{b}{a} u_2 \right)^2 + \left(\frac{\det(A)}{a} \right) u_2^2 \geq 0$$

a ako isprati bol rešavanje, i pravi rešenje vidimo, ako a_1 i a_2 su manje, normal nula.

Ima me dodatni pozitivni definitni.

Ima dodatni implicitno općenito, moram $\langle, \rangle$ je skalarni proizvod na $\mathbb{R}^2 \Rightarrow a_{11} > 0$ i $\det(A) > 0$

$$\text{Proizvod skalarni proizvod } \langle \cdot, \cdot \rangle: \langle \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \end{pmatrix} \rangle = \begin{pmatrix} 1 & 0 \end{pmatrix} \begin{pmatrix} a & b \\ b & c \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = a$$

Neka $\langle, \rangle$ je pozitivni definitni, ako $a > 0$ i tada me dodatni normalni vektor, ako $a \neq 0$. Tada $a > 0$.

Tada mi normalni je pravi implicitni:

$$a \left(u_1 + \frac{b}{a} u_2 \right)^2 + \left(\frac{\det(A)}{a} \right) u_2^2 \geq 0$$

Neka $\langle, \rangle$ je pozitivni definitni, ako po dodatni normalni vektor u masi tog last strana vidimo ako nula.

Dodatni normalni vektor $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$:

$$a \left(\frac{b}{a} + \frac{b}{a} \right)^2 + \left(\frac{\det(A)}{a} \right) 1^2 > 0$$

$$\hookrightarrow \frac{\det(A)}{a} > 0$$

Neka $a > 0$, tako je $\det(A) > 0$.